

Answer **all** questions in the answer booklet provided. Please start each question on a new page. Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

1. [Maximum mark: 27]

This question considers two possible models for the occurrence of random events in a computer game.

In a new computer game, each time a player performs an action, there is a random chance that the action will be *boosted*, meaning that it provides a benefit to the player.

The designer of this computer game is considering two possible models for when to boost an action.

In the first model, the probability that an action will be boosted is constant.

(a) Suppose the probability that an action will be boosted is 0.1 .

(i) Find the probability that the first boost occurs on the third action. [2]

(ii) Find the probability that at least one boost occurs in the first six actions. [3]

(b) Suppose the probability that an action will be boosted is p , where $0 < p < 1$.

(i) Explain why the probability that the first boost occurs on the x^{th} action is $p(1 - p)^{x-1}$. [1]

Let X be the number of actions until the first boost occurs.

(ii) Hence, write down an expression, using sigma notation, for $E(X)$ in terms of x and p . [1]

Consider the sum of an infinite geometric sequence, with first term a and common ratio r ($|r| < 1$),

$$a + ar + ar^2 + ar^3 + \dots = \frac{a}{1 - r} .$$

(c) (i) By differentiating both sides of the above equation with respect to r , find an expression for $\sum_{n=1}^{\infty} nar^{n-1}$ in terms of a and r . [4]

(ii) Hence, show that $E(X) = \frac{1}{p}$. [2]

(This question continues on the following page)

(Question 1 continued)

It can be shown that $\text{Var}(X) = \frac{1-p}{p^2}$.

- (d) Find $E(X)$ and $\text{Var}(X)$ when $p = 0.1$. [2]

In the designer's second model, the initial probability that an action is boosted is 0.2, and each time an action occurs that is not boosted, the probability that the next action is boosted increases by 0.2. After an action has been boosted, the probability resets to 0.2 for the next action.

- (e) Show that the probability that the first boost occurs on the third action is 0.288. [2]

Let Y be the number of actions until the first boost occurs.

- (f) Explain why $Y \leq 5$. [1]

The following table shows the probability distribution of Y .

y	1	2	3	4	5
$P(Y = y)$	0.2	m	0.288	n	0.0384

- (g) (i) Find the value of m and the value of n . [2]
 (ii) Show that $E(Y) = 2.5104$. [2]
 (iii) Find $\text{Var}(Y)$. [2]
- (h) (i) Use the expression given in (c)(ii) to find the value of p for which $E(X) = E(Y)$. [1]
 (ii) Find $\text{Var}(X)$ for this value of p . [1]
 (iii) Hence determine, with a reason, which model provides a more consistent experience for the player with respect to boosted actions. [1]